

KNOWLEDGE GRAPH EXTRACTION FROM UNSTRUCTURED DOCUMENTS

A PROJECT PHASE-1 REPORT

ABSTRACT

Unstructured text accounts for over 80% to 85% of enterprise and scientific data across domains such as academic literature, financial disclosures, legal frameworks, clinical records, and technical manuals. Transforming this chaotic textual repository into structured, queryable knowledge is a central challenge in Modern Information Extraction (IE) and Artificial Intelligence. This project presents a comprehensive, scalable, and modular framework for **Knowledge Graph Extraction from Unstructured Documents**, leveraging state-of-the-art Natural Language Processing (NLP), Named Entity Recognition (NER), Relation Extraction (RE), Large Language Models (LLMs), and Graph Database systems.

Phase-1 focuses on establishing the theoretical foundation, systemic architecture, data ingestion pipelines, and benchmark evaluations of fine-tuned language models (Spacy, Transformer-based architectures like RoBERTa and REBEL) versus zero/few-shot LLM prompts (GPT-4/Llama-3) for RDF triplet generation (Head Entity, Relation, Tail Entity). The system addresses fundamental bottlenecks including coreference resolution, entity disambiguation, relationship taxonomy framing, and semantic graph persistence using Neo4j. The preliminary experimental evaluation demonstrates reliable pipeline orchestration, improved precision metrics, and structured semantic reasoning across diverse textual domains

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

In the contemporary digital and information-driven era, unstructured textual data is expanding exponentially across domains such as academic literature, corporate reports, financial disclosures, legal frameworks, clinical health records, and scientific patents. Estimates indicate that over 80% to 85% of all enterprise data resides in unstructured formats. Unlike structured relational databases that enforce formal table schemas, foreign key constraints, and normalized attributes, unstructured context relies on free-form human natural language. Consequently, automated machine consumption, precise analytical querying, multi-hop reasoning, and systematic information retrieval remain fundamental challenges within Artificial Intelligence, Natural Language Processing (NLP), and Knowledge Representation (KR).

Knowledge Graphs (KGs) have emerged as a premier paradigm for transforming unstructured, unstructured documents into organized, machine-interpretable networks. By representing real-world entities (such as individuals, organizations, diseases, gene sequences, or locations) as nodes and their semantic interactions as directed, typed edges, Knowledge Graphs offer a rich, interconnected semantic topology. The core expressive unit of a Knowledge Graph is the semantic triplet:

1.2 Motivation

Traditional manual construction of domain Knowledge Graphs requires extensive human curation by subject matter experts, rendering it prohibitively expensive, slow, non-scalable, and highly susceptible to cognitive bias or subjective domain drift. Automated Information Extraction (IE) pipelines overcome these operational bottlenecks by converting raw textual streams into verified knowledge structures at scale.

Furthermore, the advent of Retrieval-Augmented Generation (RAG) in Large Language Models (LLMs) highlights the limitations of standard vector-search (dense passage retrieval). Traditional vector databases rely on chunk-level cosine similarity, which frequently fails when answering

complex queries that require multi-hop relational traversal across multiple non-contiguous pages. Integrating structured Knowledge Graphs with LLMs (GraphRAG) enables deterministic reasoning over dynamic networks, significantly mitigating model hallucinations and enhancing context-grounded response generation.

Knowledge Graphs (KGs) offer a powerful paradigm for structuring knowledge. By representing real-world entities as nodes and inter-entity associations as directed **edges**, KGs convert disparate text into a semantic network. A standard unit of a Knowledge Graph is a semantic triplet:

$$\text{Triplet} = (\text{Subject}, \text{Predicate}, \text{Object})$$

For instance, the sentence "*Albert Einstein received the Nobel Prize in Physics in 1921*" yields:

- (Albert Einstein, RECV_AWARD, Nobel Prize in Physics)
- (Nobel Prize in Physics, AWARDED_IN_YEAR, 1921)

Ext автоматизируя systemizing this conversion bridges the gap between unstructured human language and deterministic, machine-readable reasoning graphs.

1.3 Scope and Applications

The scope of this project encompasses multi-sentence, multi-page document processing across arbitrary unstructured domains (corporate reports, technical manuals, scientific research literature, and legal filings). The resulting Knowledge Graph structures directly empower downstream AI enterprise applications, including: 1) Advanced GraphRAG systems that perform structured multi-hop context retrieval; 2) Automated enterprise compliance auditing; 3) Biomedical literature mapping for drug discovery and disease association tracking; and 4) Enterprise risk intelligence mapping.

1.4 Organization of the Report

This project report is organized into thirteen comprehensive sections. Section 1 presents the Abstract. Section 2 details Chapter 1: Introduction, covering background, research motivation, scope, and structural overview. Section 3 provides an exhaustive Literature Survey tracking historical extraction methodologies, transformer architectures, and graph database storage. Section 4 defines the Problem Statement and Project Objectives. Section 5 and Section 6 analyze the Existing System limitations and Proposed System architecture, respectively. Section 7 details the Requirement Specifications across functional, non-functional, hardware, and software.

CHAPTER 2: LITERATURE SURVEY

2.1 Evolution of Knowledge Extraction Techniques

Information Extraction (IE) from natural language text has undergone major paradigmatic shifts over the past three decades. The earliest architectures in the 1990s relied on hand-crafted heuristic rules, lexicon gazetteers, and strict regular expression patterns (e.g., Hearst Patterns for hyponym extraction). While rule-based systems exhibited near-perfect precision on rigid, template-driven context, they suffered from catastrophic recall degradation when confronted with sentence structure variations, expressive domain nuances, or informal syntax.

2.3 Machine Learning and Statistical Approaches

2.2 Rule-Based and Dictionary Systems

Rule-based systems utilized shallow syntactic parsing, dependency trees, and dictionary matching algorithms to identify domain entities and pre-specified relations. The introduction of tools such as SystemT and GATE enabled scalable execution over large corpora. However, maintaining hand-crafted dictionaries and rule systems required continuous engineering effort, creating brittle software pipelines that failed to scale across dynamic multi-domain enterprise applications.

2.3 Machine Learning and Statistical Approaches

The transition toward statistical machine learning in the early 2000s introduced probabilistic sequence labeling models, notably Hidden Markov Models (HMMs), Maximum Entropy Markov Models (MEMMs), and Conditional Random Fields (CRFs). CRFs addressed the label-bias problem of MEMMs and achieved high benchmarks on sequential Named Entity Recognition (NER) tasks by optimizing global sequence likelihoods over hand-engineered features (e.g., POS tags, word shape, capitalization, and n-grams).

2.4 Deep Learning and Transformer-Based Extraction

- **BiLSTM-CRF Models (Lample et al., 2016):** Set early benchmarks in Named Entity Recognition by modeling sequential token dependencies combined with structural output constraints.
- **Transformer Encoders (Devlin et al., 2019):** BERT and its variants (RoBERTa, SpanBERT) allowed bidirectional contextual token representations, drastically improving entity boundary identification and joint entity-relation extraction models.
- **End-to-End Sequence-to-Sequence RE (Cabot & Navigli, 2021 - REBEL):** Reformulated relation extraction as an autoregressive translation task, converting context paragraphs directly into linearized triplets.

The advent of deep neural networks revolutionized Information Extraction. Recurrent Neural Networks (RNNs), specifically Bidirectional Long Short-Term Memory (BiLSTM) networks combined with CRF classification layers (BiLSTM-CRF), eliminated the need for manual feature engineering by learning contextual feature representations directly from word and character embeddings.

In 2018, Transformer architectures fundamentally transformed NLP. Pre-trained language representations such as BERT (Bidirectional Encoder Representations from Transformers), RoBERTa, and SpanBERT enabled bidirectional contextual modeling at scale. PURE (Pipeline Approach for Entity and Relation Extraction) .

To overcome two-stage pipeline error propagation, autoregressive sequence-to-sequence models emerged. REBEL (Relation Extraction By End-to-end Language generation) reformulated relation extraction as a translation task, fine-tuning BART to translate raw sentence paragraphs directly into linearized strings of RDF triplets delimited by special tokens (<triplet>, <subj>, <obj>). Most recently, Large Language Models (LLMs) such as GPT-4, Llama-3, and Mistral have demonstrated powerful zero-shot and few-shot extraction capabilities. Prompt engineering techniques combined with structured output schemas (such as Pydantic and JSON-schema) enforce strict formatting and output constraints, allowing direct extraction of normalized entities and relation predicates.

2.5 Graph Database Implementations

Storing structured knowledge in native graph databases like Neo4j enables optimized index-free adjacency traversal and flexible Cypher querying, overcoming performance bottlenecks typical of traditional relational schemas.

2.6 Comparative Analysis of Existing Literature

Author / Paper	Architecture	Key Advantages	Limitations	Target Metrics
Lample et al. (2016)	BiLSTM-CRF	Strong character-level boundary detection.	Fails to capture long-range contextual dependencies.	CoNLL-2003 F1: ~90.9%
Zhong & Chen (2020) [PURE]	Pipeline BERT Representation	Separate optimization for entity and relation representations.	Two-stage propagation error; high compute cost.	TACRED F1: ~71.5%
Cabot & Navigli (2021) [REBEL]	Seq2Seq (BART-based)	Jointly extracts multiple entities and relations in one forward pass.	Struggles on very long contexts; prone to token limits.	DocRED F1: ~51.2%
Edge et al. (2024) [GraphRAG]	LLM + Graph Clustering	Combines global thematic summarization with graph structural search.	High API latency and token cost; operational dependency on closed models.	QnA Quality & Context Recall

CHAPTER 3: PROBLEM STATEMENT

Problem Statement

Despite rapid developments in transformer-based NLP and LLMs, automated Knowledge Graph Extraction from unstructured multi-page documents suffers from key failure modes:

1. **Coreference Blindness:** Inability to link pronouns ("he", "it", "the company") back to their target entity across multi-sentence paragraphs.
2. **Cascading Error Propagation:** Errors in Named Entity Recognition (NER) propagate directly into Relation Extraction (RE).
3. **Ontological Rigidity versus Hallucination:** Traditional statistical NLP models require predefined relation taxonomies and fail on unseen domains, whereas naive zero-shot LLM extractors frequently hallucinate non-existent facts or construct inconsistent relation predicates.

There is a critical requirement for an integrated, robust, and scalable pipeline capable of extracting structured, disambiguated, and semantically verified Knowledge Graphs from raw unstructured multi-page documents.

Objectives of the Project

The overarching goal of this project is to research, design, implement, and benchmark an automated end-to-end framework for Knowledge Graph Extraction from unstructured multi-page documents. The detailed sub-objectives encompass:

1. **Document Ingestion & Artifact Cleansing:** Develop robustness against raw document structures (PDF, DOCX, HTML) by extracting clean text streams while suppressing headers, footers, sidebars, and pagination artifacts.
2. **Contextual Coreference Resolution:** Integrate neural coreference models (e.g., fastcoref, LingMess) to rewrite document chunks and resolve pronominal and nominal mentions back to canonical antecedents.
3. **Hybrid Joint Extraction Framework:** Benchmark fine-tuned autoregressive sequence-to-sequence models (REBEL) against schema-constrained LLM prompting (Llama-3/GPT-4 via Pydantic) for high-precision entity-relation extraction.

CHAPTER 4: EXISTING SYSTEM

The existing paradigms for information extraction from unstructured texts generally rely on either traditional statistical NLP pipelines or basic vector database retrieval mechanisms (Dense Passage Retrieval / RAG). Traditional systems employ multi-stage pipelines where text is processed sequentially: tokenization, POS tagging, NER boundary recognition, and supervised relation classification.

In modern enterprise environments, standard Retrieval-Augmented Generation (RAG) relies on chunking documents into static token windows and computing dense vector embeddings (e.g., cosine similarity search via FAISS/ChromaDB). While effective for direct keyword or semantic passage retrieval, these systems exhibit fundamental limitations:

1. **Lack of Multi-Hop Reasoning:** Vector similarity fails when answering complex queries requiring multi-step relational links across distant document pages.
2. **Pipeline Error Cascading:** Traditional NER-RE pipelines suffer from propagated classification errors from early stages into relation prediction.
3. **Entity Duplication:** Standard vector search cannot resolve entity surface variations ("IBM", "International Business Machines"), leading to fragmented search indexes.

CHAPTER 5: PROPOSED SYSTEM

The proposed system introduces an end-to-end, automated Knowledge Graph Extraction framework designed to overcome coreference blindness, entity fragmentation, and relation hallucination. The architecture integrates neural coreference resolution, fine-tuned seq2seq / LLM extraction engines, vector similarity entity canonicalization, and native graph persistence in Neo4j. Key highlights and workflow of the proposed architecture include:

1. **Robust Text Preprocessing:** Automated document parsing with sliding-window chunking (512 tokens with 64-token overlap) and noise removal.
2. **Neural Coreference Layer:** Replacing pronominal and nominal mentions with canonical antecedents prior to extraction.
3. **Hybrid Extraction Engine:** Combining high-speed seq2seq translation (REBEL BART-large) with schema-constrained LLM prompting (Llama-3/GPT-4 via Pydantic) to extract verified RDF triplets.
4. **Vector-Based Entity Canonicalization:** Applying Levenshtein string distance and embedding cosine similarity to merge redundant entity nodes.
5. **GraphRAG Integration:** Persisting structured graph networks to Neo4j, enabling deterministic multi-hop querying and hybrid graph-vector RAG.

The system methodology is structured as a sequential multi-stage computational pipeline. Raw multi-page document inputs undergo artifact filtering and sliding-window chunking. The chunked streams pass into a coreference resolution module, replacing pronominal mentions with canonical antecedents. Resolved text chunks are processed through joint entity-relation extractors (REBEL seq2seq or schema-constrained LLMs). Extracted subject-predicate-object triplets are normalized through embedding cosine similarity to resolve entity duplicates, and compiled into Cypher transaction batches for atomic Neo4j graph storage.